$$\lim_{x\to 0^+}(120-15x) \le \lim_{x\to 0^+} x\left[\left\lfloor\frac{1}{x}\right\rfloor+\left\lfloor\frac{2}{x}\right\rfloor+\cdots+\left\lfloor\frac{15}{x}\right\rfloor\right] \le \lim_{x\to 0^+}(120+15x)$$

$$120 \le \lim_{x\to 0^+} x\left[\left\lfloor\frac{1}{x}\right\rfloor+\left\lfloor\frac{2}{x}\right\rfloor+\cdots+\left\lfloor\frac{15}{x}\right\rfloor\right] \le 120$$

$$\lim_{x\to 0^+} x\left[\left\lfloor\frac{1}{x}\right\rfloor+\left\lfloor\frac{2}{x}\right\rfloor+\cdots+\left\lfloor\frac{15}{x}\right\rfloor\right]=120\,.$$

9.2.9 Two special Trigonometrical limits

Result 9.1

(a) $\lim_{\theta\to 0}\frac{\sin\theta}{\theta}=1$ (b) $\lim_{\theta\to 0}\frac{1-\cos\theta}{\theta}=0$.

Proof

We use a circular sector to prove the result.

Consider the circle with centre (0, 0) and radius 1. Any point on this circle is $P(\cos\theta, \sin\theta)$.

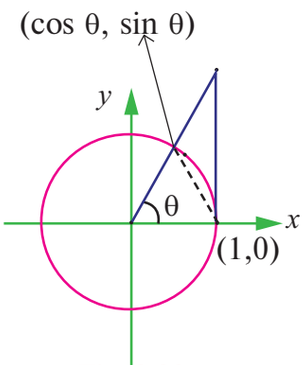
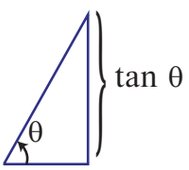


Fig. 9.28



Area of triangle $\frac{\tan\theta}{2}$

Fig. 9.29



Area of sector $\frac{\theta}{2}$

Fig. 9.30



Area of triangle $\frac{\sin\theta}{2}$

Fig. 9.31

By area property $\frac{\tan\theta}{2} \ge \frac{\theta}{2} \ge \frac{\sin\theta}{2}$.

Multiplying each expression by $\frac{2}{\sin\theta}$ produces $\frac{1}{\cos\theta} \ge \frac{\theta}{\sin\theta} \ge 1$ and taking reciprocals

$\cos\theta \le \frac{\sin\theta}{\theta} \le 1$.

Because $\cos(-\theta) = \cos\theta$ and $\frac{\sin(-\theta)}{-\theta} = \frac{\sin\theta}{\theta}$ one can conclude that this inequality is valid for all non-zero θ in the open interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

We know that $\lim_{\theta\to 0}\cos\theta = 1; \lim_{\theta\to 0}(1) = 1$ and applying Sandwich theorem we get $\lim_{\theta\to 0}\frac{\sin\theta}{\theta}=1$. ■

(b) $\lim_{\theta\to 0}\frac{1-\cos\theta}{\theta}=0$.

$$1-\cos\theta = 2\sin^2\frac{\theta}{2}$$

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